Circuit investigation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements to test a circuit model safely.

SUCCESS CRITERIA - I can...

1. Draw and predict before powering a circuit.
2. Place ammeters in series and voltmeters in parallel.
3. Compare measured and predicted values using evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ammeter” and “voltmeter” to explain your model or result. Sentence starters are optional.

ammeter: an instrument placed in series to measure current

voltmeter: an instrument placed across two points to measure voltage

trial: one run or repetition of a test

01 Warm-up: bring back an earlier idea

An ideal series loop has supply 15 V. One component drops 3 V.
Find the other component’s drop.

Word help: series = connected along one unbranched current path

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

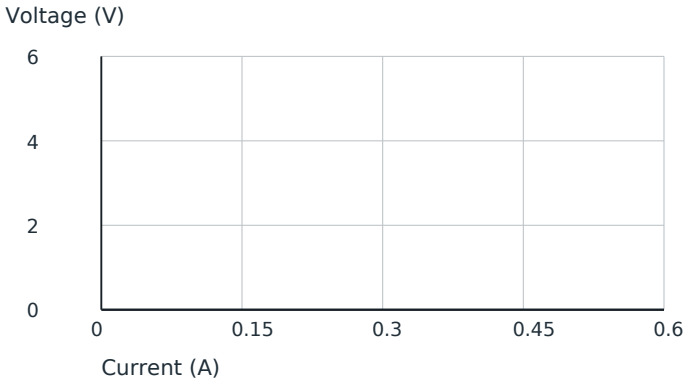
02 Worked example: explain the model

Reference: V = IR.
Use approved low voltage. Ammeter in series; voltmeter across a component. Disconnect before rewiring; never put an ammeter across the supply.

Worked reasoning: Supplied measurements show V=2, 4, 6 V and I=0.2, 0.4, 0.6 A. Each V/I is 10 ohms. This supports an ohmic model over the measured range.

Your turn: Plot V against I and mark the slope.
State why the data do not prove the model at every possible temperature or voltage.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

On a resistor circuit sketch, add an ammeter to measure loop current and a voltmeter to measure the resistor’s drop.

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

Before powering a teacher-approved circuit, predict current for a 6 V supply and 12 Ω resistor. Then use the table to record actual voltage, current and observations during the test. Keep measured entries blank until you take measurements.

Sentence starter: I predict … . My model supports this because … .

Voltage (V)	Current (A)	Observation

Marzano check - circle: ① ② ③

05 Explain your reasoning

A reading disagrees with your prediction. List two checks before replacing the model or deleting data.

Sentence starter: I predict … . My model supports this because … .

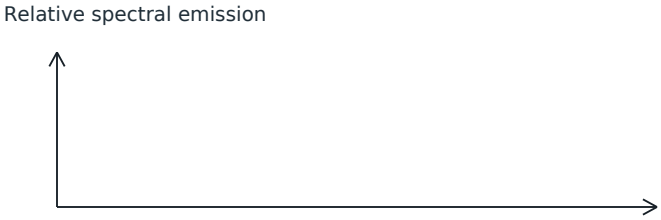
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An ideal blackbody’s kelvin temperature becomes 4 times as large. How does its peak wavelength change? **Show** how the thermal peak changes.

Word help: blackbody = an ideal absorber/emitter used in a thermal-radiation model

Sentence starter: I notice … . This supports … because … .



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Before the first powered test, have the teacher check the circuit connections and meter placement.
Mark the checked locations on the diagram and record one actual comment or change. Do not invent feedback.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A later reading is 8 V and 0.5 A. Does it fit the same 10 ohm model?
Name two checks before declaring the physical law wrong.

Sentence starter: The step I would change is ... . The evidence is ... .

Quantity	Lower bound	Central value	Upper bound
Voltage (V)	____	____	____
Current (A)	____	____	____
Resistance (Ω)	____	____	____

Marzano check - circle: 1 2 3